Prathmesh Sayal

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Summary

I have always been driven by curiosity about how things work and a desire to understand systems by taking them apart and building them again. Building my first custom PC in 10th grade led to years of experimentation with overclocking, virtualisation, and hardware, eventually developing into an interest in computer science and cybersecurity. This curiosity has since evolved into a broader research interest in artificial intelligence, particularly how AI can be made more trustworthy, interpretable, and secure. Through research projects, publications, internships at ISRO and HPE, and competitive technical projects, I have sought to explore these questions while developing the ability to turn ideas into working systems.

Research Interests

Trustworthy & Interpretable AI · AI Safety · Deep Learning · Computer Vision · Machine Learning · AI Security

Education

Ramaiah Institute of Technology (RIT), Bengaluru, India — Aug 2023 – May 2027
B.E. in Computer Science and Engineering, Specialization in Cybersecurity — **CGPA: 9.52/10.00**

Relevant Coursework: Statistics & Linear Algebra, Discrete Mathematics, Advanced Calculus, Applied Graph Theory, Database Management Systems, Cryptography, Intro to AI/ML, Natural Language Processing, Quantum Computing, Data Structures, Algorithms, Computer Architecture, Operating Systems

Internship Experience

Indian Space Research Organisation, Trainee, 1 among 5 nationally — Jul 2026 – Aug 2026
- Co-developed a native C17 object detector fine-tuned from PicoDet-S, reducing fine-tuning to under 10 seconds on resource-constrained CPU across generalized target datasets
- Evaluated FP8 (E4M3) arithmetic under a static execution schedule for planned FPGA deployment, quantifying 15% accuracy trade-off against FP32 and 50x reduction in memory usage

Fidelity Investments, Summer Intern — May 2026 – Jul 2026
- Built a reusable automation framework for extraction, validation, persistence workflows across data pipelines
- Automated validation of extracted data from Azure Blob Storage against other source records

Hewlett Packard Enterprise, Project Intern — Apr 2025 – Jan 2026
- Developed a confidential-computing prototype for protection of data in-use, building a zero-trust telemetry aggregation pipeline that sent encrypted metrics & enabled federated learning across them with 78% accuracy
- Implemented continuous Trusted Platform Module(TPM) remote attestation with a centralized verification service and flagging compromised nodes within seconds

Indian Space Research Organisation, Trainee — Aug 2025 – Sep 2025
- Developed an AI pipeline for processing and interpreting satellite imagery, improving object recognition for autonomous navigation while supporting sub-meter precision and inference latency below 20 ms.
- Prototyped a machine-learning algorithm for docking-port target detection in support of Indian Space Station Programme

Research Projects

CortexMD: Explainable Multimodal Clinical AI (Team Lead) — GitHub
- Led development of a multimodal medical diagnosis system with reconstructing full 3D anatomical models from 2D scans using **NVIDIA CLARA**, with real-time risk monitoring
- Addressed the black-box trust problem in clinical AI by building an **explainable-AI** pipeline combining first-order logic, knowledge graphs, and confidence-based verification

Fence.AI: Deepfake Protection and Explainable Detections — GitHub
- Devised a tool that protects ordinary images from deepfake generation by injecting adversarial perturbations into the image's **latent space**, hallucinating image-gen models before they can produce a deepfake
- Used **Local Interpretable Model-Agnostic Explanations (LIME)** to surface the reasoning behind the deepfake detection model's inference, giving feature-level breakdown of why an image was flagged

Publications & Manuscripts

- **Prathmesh Sayal**, Kamal Karteek U, Praroop Sharma, and Rakesh Kalshetty. "Quantum Powered Data Analysis." *Proceedings of the International Conference on Emerging Technologies 2024 (ICEMTEC 2024)*, India, 28–29 May 2024 *(Published)*
- **Prathmesh Sayal**, Kshiraja Nelapati. "HRT-LI: Certified Rank Transport for Dynamic Learned Index over Hierarchical String Keys." *Journal of Big Data*, Springer 2026 *(Preprint)*
- Kshiraja Nelapati, Meeradevi A. K., and **Prathmesh Sayal**. "Explainable Hybrid Liquid AI for Robust Neurodiagnostics in Low-Resource Areas." *CSIT*, Indonesia, 2026 *(Submitted)*
- Vishalakshi Prabhu H., Shankaramma, **Prathmesh Sayal**, Ishan Chitkarsh, Dhanush J., and Bhaskar Datta Paramkusam. "Smart Honeypot for Threat Detection in IoMT Networks." *Discover Internet of Things*, Springer, 2026 *(With Editor)*
- Meeradevi, Vineet Sadarangani, Darshan Bankapure, Hrishikesh Haritas, and **Prathmesh Sayal**. "GenAI Framework for Expediting the Search for New Therapeutics Using Generative Adversarial Networks." *JCOMSS*, 2026 *(Accepted)*

Achievements

10x Hackathons Winner · 6x CTFs Winner

- **Winner, JIT Hack 2024** – Built GuardAI, the initial prototype of Fence.AI, extending PhotoGuard (MIT Madry Lab).
- **Winner, CIDECODE PES 2025** – Fence.AI recognised by Data Security Council of India
- **Runner-Up, Smart India Hackathon(SIH) 2024** – Facilitated by the Government of India and Government of Rajasthan; the solution is now in production.
- Also 1st at **HPE Think-a-thon**, 2nd at **PragyanCTF**, and top 70 of 10,000+ teams at **PicoCTF** (conducted by CMU).

Societies

- **secuRIT** – Founding Member — Oct 2024 – Dec 2025
- **Google Developer Group, RIT** – Lead — Dec 2024 – Present
- **IEEE Computational Intelligence Society** – EXECOM — Jan 2025 – Jan 2026
- **National Service Scheme** – Member — Oct 2025 – Present

Technical Skills

Programming	Python, C, C++, Java
Machine Learning	PyTorch, TensorFlow, Keras, Generative AI, Explainable AI, Computer Vision
Systems & Cloud	Docker, Kubernetes, AWS, Linux/Unix, Git, GitHub
Data & Backend	SQL, MongoDB, Node.js, REST APIs

Extracurricular Engagement

- **Led** student technical initiatives; organised SIH 2025 mentorship and **judging**, working with participants, mentors, and fellow organisers
- **Mentored** peers through beginner ethical-hacking workshops, helping students build practical skills and confidence
- **Collaborated** with the executive team to coordinate speakers, logistics, and student participation
- **Volunteered** for a blood donation camp as a National Service Scheme member.